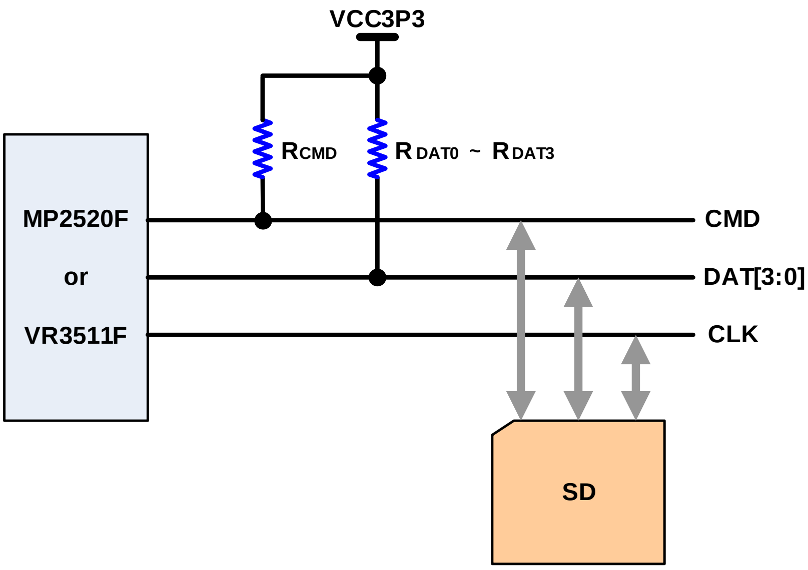
NOTICE> The R_{CMD} resistor value to increase SD/MMC card compatibility (10Kohm)

1. SD

R_{DAT} and R_{CMD} are pull-up resistors protecting the CMD and the DAT line against bus floating when no card is inserted or when all card drivers are in a hi-impedance mode. The extended DAT lines(DAT1 ~ DAT3) are input on power up. They start to operate as DAT lines after the SET_BUS_WIDTH command. It is the responsibility of the MP2520F(or VR3511F) designer to connect external pull-up resistors(R_{DAT0}~R_{DAT3}) to all data lines even if only DAT0 is to be used. Otherwise, non-expected high current consumption may occur due to the floating inputs of the extended DAT lines. For more information, refer to the SD specification.



<Figure 1. SD Interface>

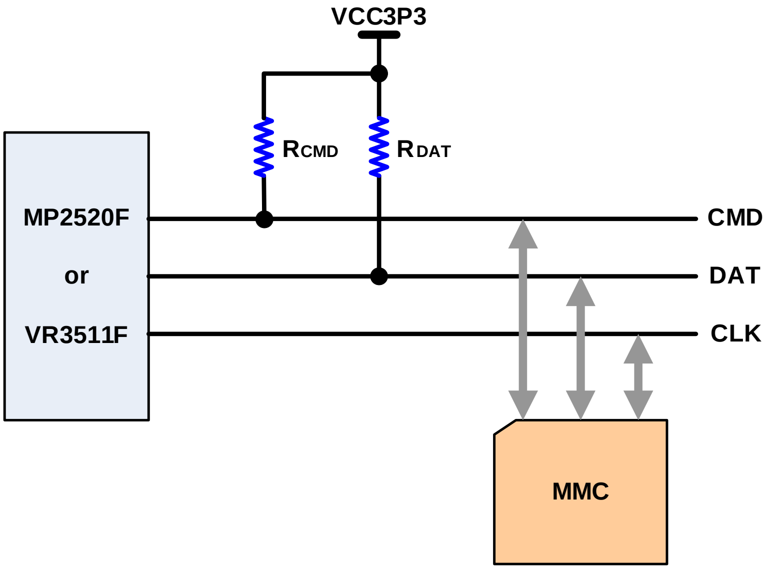
In the SD specification, the recommended value of R_{DAT} and R_{CMD} is 50Kohm(+/- 20Kohm). <Table 1> shows the recommended value of Magiceyes.

Resistor	Value
R_{CMD}	10K ohm
R_{DAT0}	47K ohm
R_{DAT1}	47K ohm
R_{DAT2}	47K ohm
R_{DAT3}	47K ohm

<Table 1. The pull-up resistors value for SD card>

2. MMC

R_{DAT} and R_{CMD} are pull-up resistors protecting the CMD and the DAT line against bus floating when no card is inserted or when all card drivers are in a high-impedance mode. In addition to that, unlike SD's CMD, MMC's CMD is driven in open-drain mode so pull-up resistor's value is very important to increase card compatibility.



<Figure 2. MMC Interface>

The R_{CMD} resistor value must be customized within the range given in the MMC specification according to such conditions as the bus operating frequency, number of MMC slots, and the load capacitance of the cards. <Table 2> shows the recommended pull-up resistor value for R_{CMD} and R_{DAT} in the MMC specification.

Resistor	Value
R_{CMD}	4.7K ~ 100K ohm
R_{DAT}	50K ~ 100K ohm

<Table 2. The recommended resistor value of the MMC specification>

Magiceyes have tested to increase the MMC card compatibility and <Table 3> shows the result.

Resistor	Value
R_{CMD}	10K ohm
R_{DAT}	47K ohm

<Table 3. The pull-up resistor value for MMC card>

Magiceyes strongly recommend <Table1> and <Table3> value to customer.